From variational Monte Carlo to superfluid hydrodynamics in microwave-shielded dipolar Bose gases

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We show that, within the manifold where the Jastrow pair factor is frozen to its ground-state form, time-dependent variational Monte Carlo (tVMC) based on a Jastrow–Feenberg wave function maps exactly onto irrotational superfluid hydrodynamics. The theory is then closed by an equation of state extracted from a single variational Monte Carlo calculation of the uniform system; within the local-density approximation, this yields a generalized Gross–Pitaevskii equation with a density-dependent chemical potential $\mu(\rho)$. For a trapped microwave-shielded dipolar Bose gas, the reduced dynamics reproduces the full tVMC breathing oscillations for $N = 8$–128 with below-1% frequency error and $\sim 5\%$ amplitude error, while lowering the computational cost by more than an order of magnitude. Our results establish a first-principles route from correlated many-body wave functions to predictive hydrodynamics for molecular gases, quantum droplets, and supersolids.

Introduction.—Microwave-shielded polar molecules have emerged as a leading platform for dipolar quantum matter. Recent experiments have achieved Bose–Einstein condensation [1, 2], observed shielded collisions [3, 4], and probed few- and many-body phenomena [5–7]. With permanent dipole moments orders of magnitude larger than those of magnetic atoms [8, 9], shielded molecules are expected to exhibit rich collective behavior—from quantum droplets [10, 11] to supersolid-like phases—in a widely tunable setting. Collective oscillations and expansion dynamics serve as primary experimental probes of the underlying interactions in trapped BECs [12–14]; for shielded molecular condensates such measurements remain an open frontier, making first-principles predictions especially timely.

For magnetic-atom dipolar BECs, the extended Gross–Pitaevskii equation (eGPE)—incorporating the non-local dipolar mean field and a perturbative Lee–Huang–Yang (LHY) correction [15]—has successfully described quantum droplets and collective excitations [8, 11]. This framework, however, relies on a Born-approximation coupling constant and treats quantum fluctuations perturbatively; indeed, quantum Monte Carlo calculations have revealed significant deviations from eGPE predictions even for magnetic atoms [16]. For microwave-shielded molecules the situation is more challenging: the strong short-range repulsion ($\sim C_6/r^6$) produces a deep correlation hole that cannot be captured perturbatively, and without it the bare interaction integral diverges. Time-dependent variational Monte Carlo (tVMC) [17–19] can propagate a correlated many-body wave function in real time, but its stochastic sampling at every time step severely limits accessible system sizes and evolution times. What is missing is a controlled coarse-graining that retains the microscopic content of tVMC while recovering the efficiency of hydrodynamic evolution.

Here we establish a first-principles reduction from correlated tVMC to macroscopic hydrodynamics. The key physical ingredient is a separation of scales: short-range pair correlations relax much faster than the global density profile. Under this condition, freezing the two-body Jastrow factor u_2 makes the time-dependent variational principle map exactly onto irrotational superfluid hydrodynamics. We verify this scenario from full tVMC, then close the theory with an equation of state extracted from a single uniform-system VMC calculation within the local-density approximation, yielding a generalized GPE. The reduced theory reproduces the full tVMC breathing dynamics for $N = 8$ and $N = 128$, while extending directly to experimentally relevant large-N regimes beyond direct tVMC.

Model and variational dynamics.—We consider N bosons in a cylindrically symmetric harmonic trap with Hamiltonian

$$\hat{H}(t) = \sum_{i=1}^{N}\left[-\tfrac{1}{2}\nabla_i^2 + V_{\rm tr}(\mathbf{r}_i, t)\right] + \sum_{i<j} v(\mathbf{r}_{ij}), \qquad (1)$$

where $V_{\rm tr} = \frac{1}{2}[\omega_\rho(t)^2\rho^2 + \omega_z^2 z^2]$ (natural units $\hbar = m = 1$) and the two-body interaction models microwave-shielded dipolar molecules:

$$v(\mathbf{r}) = \frac{C_3}{r^3}(3\cos^2\theta - 1) + \frac{C_6}{r^6}(\sin^4\theta + w_{12}\sin^2\theta\cos^2\theta). \quad (2)$$

This effective potential, derived from Floquet scattering theory for dual-microwave-dressed molecules [20], is parametrized by coefficients C_3, C_6, and w_{12} fixed by the microwave field configuration. For the representative case studied here we choose $C_3 = 10^{-3}$, $C_6 = 10^{-6}$, and $w_{12} = 2.0$ in oscillator units, initialize the system in the isotropic trap $\omega_\rho = \omega_z = 1$, and excite radial breathing dynamics with a quench $\omega_\rho : 1 \to 2$ at $t = 0$.

The many-body wavefunction is a truncated Jastrow–Feenberg [21, 22] ansatz:

$$\Psi(\mathbf{X},t) = \exp\Big[\sum_i u_1(\mathbf{r}_i,t) + \sum_{i<j} u_2(r_{ij},\hat{\mathbf{r}}_{ij})\Big], \qquad (3)$$

where the complex one-body field $u_1(\mathbf{r},t)$ controls the evolving density and phase, while u_2 encodes pair correlations. In cylindrical geometry we use a separable parametrization for u_1, and expand u_2 in partial waves as

$$u_2(r,\cos\theta) = u_{\rm cusp}(r) + \sum_{\ell=0,2,4} f_\ell(r)\, P_\ell(\cos\theta), \qquad (4)$$

where $u_{\rm cusp}(r)$ enforces the short-range scattering boundary condition and each f_ℓ is represented by Gaussian basis functions [23]. For the present parameters the $\ell = 2$ anisotropy is weak and the $\ell = 4$ channel is negligible, so the pair correlation is close to isotropic.

Figure 1(a) characterizes the optimized pair correlations. The lowest-order hypernetted-chain estimate $g_2^{\rm HNC}(r,\theta) = e^{2u_2(r,\cos\theta)}$ [24] shows a short-range correlation hole produced by the shielding repulsion, with only weak angular dependence.

The TDVP [25, 26] projects the Schrödinger equation onto the variational tangent space. Writing $O_\alpha = \partial_{\theta_\alpha} \ln\Psi$ and $E_L = (\hat{H}\Psi)/\Psi$, the equations of motion are [17]

$$\sum_\beta S_{\alpha\beta}\,\dot{\theta}_\beta = -i\,F_\alpha, \qquad (5)$$

where the overlap matrix and force vector are

$$S_{\alpha\beta} = \langle \Delta O_\alpha^* \,\Delta O_\beta\rangle_t, \qquad (6)$$
$$F_\alpha = \langle \Delta O_\alpha^* \,\Delta E_L\rangle_t, \qquad (7)$$

with $\Delta A = A - \langle A\rangle_t$ denoting connected fluctuations. Full Monte Carlo and convergence details are given in Ref. [23].

Frozen-u_2 reduction.—The reduction is motivated by a clear separation of scales: the pair correlation u_2 is governed by the interaction potential $v(r)$ and adjusts on the fast time scale $\tau_{\rm pair} \sim r_0^2$ set by the range r_0 of the repulsive shield, whereas the trap quench drives density redistribution on the much slower trap time scale $\sim 1/\omega$. This hierarchy—analogous to the Born–Oppenheimer separation between fast electronic and slow nuclear degrees of freedom [27]—suggests that the pair-correlation sector remains adiabatically pinned to its ground-state form while only the one-body field u_1 evolves. The comparison between full tVMC and this frozen-u_2 approximation confirms the picture quantitatively: Fig. 1(b) shows that the frozen-u_2 evolution closely tracks the full tVMC dynamics for $N = 128$, validating the restriction to fixed u_2.

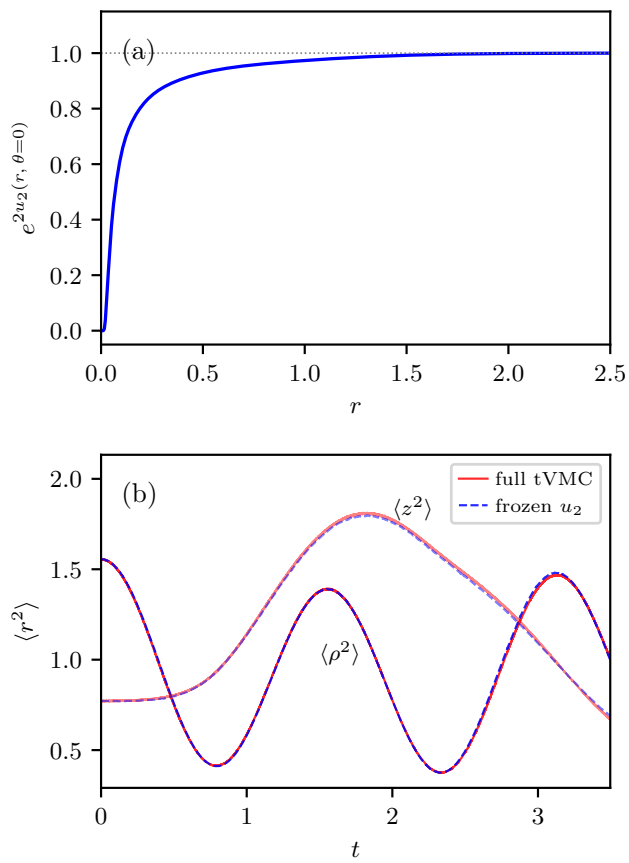

FIG. 1. (a) Lowest-order HNC pair correlation $e^{2u_2(r,\theta=0)}$ for the N=8 ground state obtained by imaginary-time tVMC in the initial isotropic trap $\omega_\rho = \omega_z = 1$, showing a correlation hole at short range due to the shielding repulsion. (b) Breathing dynamics after the quench $\omega_\rho : 1 \to 2$ for N=128: $\langle\rho^2\rangle(t)$ and $\langle z^2\rangle(t)$ from full tVMC (red) and frozen-u_2 tVMC (blue dashed), validating the frozen-u_2 approximation.

When the variational degree of freedom is the continuous field $u_1(\mathbf{r},t)$, the TDVP logarithmic derivative becomes the microscopic density operator:

$$O(\mathbf{r};\mathbf{X}) = \frac{\delta \ln\Psi}{\delta u_1(\mathbf{r})} = \sum_i \delta^{(3)}(\mathbf{r}-\mathbf{r}_i) = \hat{\rho}(\mathbf{r}). \qquad (8)$$

Because u_2 is real and frozen, the many-body phase is purely one-body and therefore irrotational, $\mathbf{v} = \nabla(\mathrm{Im}\, u_1)$; no backflow terms arise. The resulting potential flow precludes vortex formation, so the approximation is tailored to collective modes and quench dynamics that remain irrotational.

Decomposing $u_1 = a + i\theta$ (real amplitude and phase), the TDVP action reduces to the standard hydrodynamic form [23]. The reduction is exact within this frozen-u_2

manifold and yields the continuity and Euler equations:

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0, \tag{9}$$

$$\partial_t \mathbf{v} + \nabla\Big(\tfrac{1}{2} v^2 + V_{\rm tr} + \mu_{u_2}[\rho]\Big) = 0, \tag{10}$$

where $\mu_{u_2}[\rho] = \delta E_{\rm int}[\rho; u_2]/\delta\rho$ is the generally nonlocal chemical-potential functional determined by the frozen correlations.

To obtain a closed local evolution equation, we next invoke the local-density approximation (LDA). The internal energy then takes the form $E_{\rm int} = \int \rho\, \varepsilon(\rho)\, d^3 r$, where $\varepsilon(\rho)$ is the energy per particle of the *uniform* system at density ρ with the same frozen u_2. This approximation is valid when the density varies slowly on the scale of the pair-correlation range, $|\nabla\rho|\, r_0/\rho \ll 1$, a condition well satisfied in the bulk of the trapped cloud. The chemical potential becomes a local function,

$$\mu(\rho) = \frac{d[\rho\varepsilon(\rho)]}{d\rho} = \varepsilon + \rho\varepsilon'. \tag{11}$$

The hydrodynamic theory therefore closes in GP form, except that the usual contact term $g\rho$ is replaced by the VMC-derived function $\mu(\rho)$.

The near-stationarity of the pair correlations during the quench is visualized directly in Fig. 2, which shows the change in the lowest-order HNC pair correlation Δe^{2u_2} at four time snapshots. The relative variation remains below $\sim 3\%$ throughout the breathing cycle, even as the one-body density undergoes order-unity redistribution, confirming that u_2 is effectively frozen on the dynamical time scale.

Equation of state and generalized GPE.—We extract $\varepsilon(\rho)$ from a one-time VMC calculation on the uniform system (periodic boundary conditions) with u_2 frozen at its trapped ground-state values (computational details are given in [23]). Over the relevant density range, we fit the results to a polynomial

$$\varepsilon(\rho) = a_1\rho + a_2\rho^2 + \cdots, \tag{12}$$

yielding the chemical potential $\mu(\rho) = 2a_1\rho + 3a_2\rho^2 + \cdots$.

The striking finding is that $\mu(\rho)$ is nearly linear over the density range relevant to the trapped dynamics (Fig. 3), so the EOS is dominated by the leading term $\mu \approx 2a_1\rho$. This has the same functional form as the standard GPE nonlinearity $g|\psi|^2$, with a *renormalized coupling* $g_{\rm eff} = 2a_1$. Physically, the frozen pair correlations—including the short-range correlation hole visible in Fig. 1(a)—regularize the bare interaction integral

$$\varepsilon_{\rm int}(\rho) = \frac{\rho}{2}\int g_2(\mathbf{r};\rho)\, v(\mathbf{r})\, d^3 r, \tag{13}$$

which would diverge for the bare C_3/r^3 potential with $g_2 = 1$. The non-trivial pair structure encoded in u_2 renders this integral finite and maps the realistic anisotropic interaction onto a single effective coupling $g_{\rm eff}$.

This construction differs from phenomenological GP theories in two respects. First, $g_{\rm eff}$ is fixed by the correlated many-body wave function rather than by a scattering-length approximation; for the present parameters we obtain $a_1 \approx 0.134$ from $N = 32$ particles in the periodic simulation cell, converged to within 3% of the $N \to \infty$ extrapolation $a_1^\infty \approx 0.137$ [23]. Second, the small quadratic correction $a_2\rho^2$ captures the residual density dependence of the pair correlations—an effective three-body contribution that emerges naturally from the VMC-derived EOS rather than being inserted ad hoc.

Unlike the eGPE [11, 15], which treats the dipolar interaction through a non-local mean-field kernel supplemented by a perturbative LHY correction, the present approach folds the full interaction—including all non-perturbative pair correlations—into a local function $\mu(\rho)$, extracted once from the uniform-system VMC. This is both computationally simpler (no long-range convolutions at each time step) and microscopically more faithful. The near-linearity of $\mu(\rho)$ is characteristic of the weak-interaction regime ($C_3 = 10^{-3}$); for stronger dipolar coupling, the higher-order polynomial terms become essential, and the framework accommodates this naturally through the full $\mu(\rho)$ nonlinearity in Eq. (14).

The resulting generalized GPE,

$$i\partial_t\psi = \Big[-\tfrac{1}{2}\nabla^2 + V_{\rm tr} + \mu(|\psi|^2)\Big]\psi, \tag{14}$$

thus takes the standard GP form with the contact coupling replaced by the VMC-derived $\mu(\rho)$; the near-linearity of the EOS explains why a mean-field-like equation captures the correlated dynamics so accurately.

GPE solver.—We solve Eq. (14) where $\psi = \sqrt{\rho}\, e^{i\theta}$ and the Laplacian provides both quantum pressure and flow kinetic energy. The equation is discretized on a cylindrical (r, z) grid exploiting azimuthal symmetry, evaluated with a fourth-order finite-difference Laplacian and time-integrated using a Runge–Kutta scheme under a dispersive CFL condition [23].

The initial state is obtained by imaginary-time propagation to self-consistency: $\psi \to \psi - d\tau\, \hat{H}\psi$, normalized to N particles at each step, until the chemical potential converges to $\delta\mu < 10^{-8}$.

Validation against tVMC.—We benchmark the generalized GPE against frozen-u_2 tVMC at both weak and moderate interaction strength. For $N = 8$, the GPE ground state reproduces the tVMC moments to $\sim 1\%$ and the energy to 0.7% (Table S1 [23]). The breathing frequency is $\omega_{\rm br} = 4.19 \approx 2\omega_\rho$ in both methods, and the oscillation amplitudes of $\langle\rho^2\rangle$ and $\langle z^2\rangle$ agree to within $\sim$1%.

For $N = 128$, where the interaction energy reaches $\sim 21\%$ of the total, the same EOS-based GPE remains accurate. The ground-state moments agree with tVMC

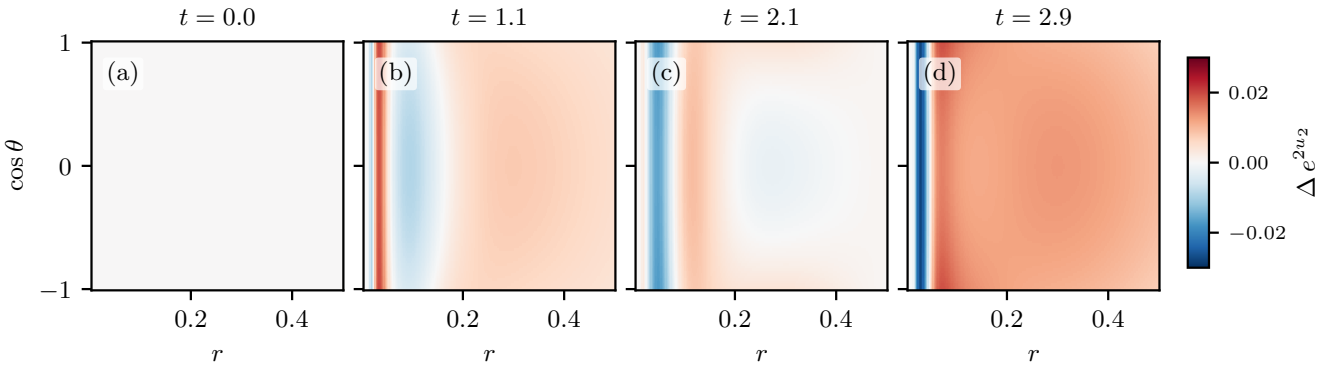

FIG. 2. Change in the lowest-order HNC pair correlation, $\Delta e^{2u_2} = e^{2u_2(t)} - e^{2u_2(0)}$, during breathing dynamics for N=128. The pair correlations remain nearly frozen ($|\Delta e^{2u_2}| \lesssim 0.03$) while the one-body density undergoes order-unity changes.

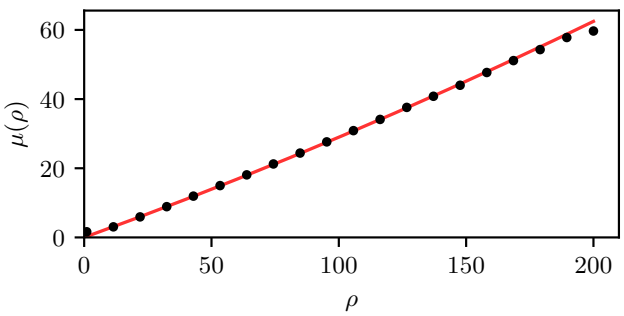

FIG. 3. Chemical potential $\mu(\rho) = d(\rho\varepsilon)/d\rho$ from uniform-system VMC (circles) for C_3=10^{-3}. The line shows the polynomial fit $\mu = 2a_1\rho + 3a_2\rho^2$ used in the GPE. The near-linearity reflects that the frozen pair correlations renormalize the dipolar interaction into an effective contact coupling $g_{\rm eff} = 2a_1$.

to 3.5% for $\langle\rho^2\rangle$ and 4.2% for $\langle z^2\rangle$, while the total energy differs by only 0.4%. The few-percent offset in moments is consistent with a residual LDA error at higher density. The dynamical benchmark in Fig. 4(a) is likewise preserved: both approaches give $\omega_{\rm br} = 4.19$, and the $\langle\rho^2\rangle(t)$ amplitude agrees to about 5%. Notably, although the quench acts only on ω_ρ, the axial moment $\langle z^2\rangle(t)$ also oscillates—driven by nonlinear radial–axial coupling through the interaction energy—and the GPE captures this cross-coupling quantitatively [Fig. 4(a)].

The computational gain is substantial. For N = 128, the GPE evolution, including imaginary-time state preparation, takes $\sim$ 70 minutes on a single CPU core, compared with $\sim$ 6 hours on a modern GPU for the full tVMC.

Large-N prediction.—Having benchmarked the reduction, we use it in a regime inaccessible to direct tVMC. With the same interaction ($C_3 = 10^{-3}$), the generalized GPE predicts sustained breathing oscillations for N = 8192 over six trap periods [Fig. 4(b)], with excellent energy conservation ($|\delta E/E| < 2.5 \times 10^{-4}$).

The N = 8192 system is qualitatively more strongly interacting than the benchmarked cases: the interaction energy comprises $\sim$ 38% of the total (compared with 3.5% for N = 8 and 21% for N = 128), and the peak density reaches $\rho_0 \approx 43$. Despite this tenfold variation in interaction fraction, the breathing frequency remains $\omega_{\rm br} \approx 4.19$ from N = 8 to N = 8192. This robustness is a direct consequence of the near-linearity of $\mu(\rho)$: for a strictly linear chemical potential, the scaling ansatz [28, 29] predicts a breathing frequency that depends only on the trap geometry and is independent of the coupling strength. The quench also drives pronounced axial oscillations: even though ω_z is unchanged, $\langle z^2\rangle$ oscillates with a relative amplitude of $\sim$ 130% [Fig. 4(b)], reflecting strong radial–axial coupling mediated by the interaction at large N.

This particle number falls within the range achieved in recent molecular BEC experiments ($N \sim 10^3$–10^4) [1, 2]. Consequently, the predicted breathing dynamics is directly measurable via experiments, providing a stringent test of the VMC-derived equation of state.

Discussion.—The success of the frozen-u_2 hydrodynamic reduction rests on two pillars: (i) a separation of time scales—pair correlations equilibrate much faster than the macroscopic density redistribution, keeping u_2 nearly stationary—and (ii) the fact that integrating out these correlations yields an EOS dominated by a single renormalized coupling $g_{\rm eff}$.

The near-linearity of $\mu(\rho)$ explains, at a deeper level, why a GPE-like equation captures the correlated dynamics so faithfully even though the microscopic pair correlation $g_2(r) = e^{2u_2}$ differs markedly from the mean-field value $g_2 = 1$. For the hydrodynamic evolution only the functional form of $\mu(\rho)$ matters, not the microscopic origin of its coefficients: once the pair correlations have been integrated into $g_{\rm eff}$, the dynamics is insensitive to the detailed shape of g_2. The value of $g_{\rm eff}$, however, *cannot* be obtained from a mean-field ($g_2 = 1$)

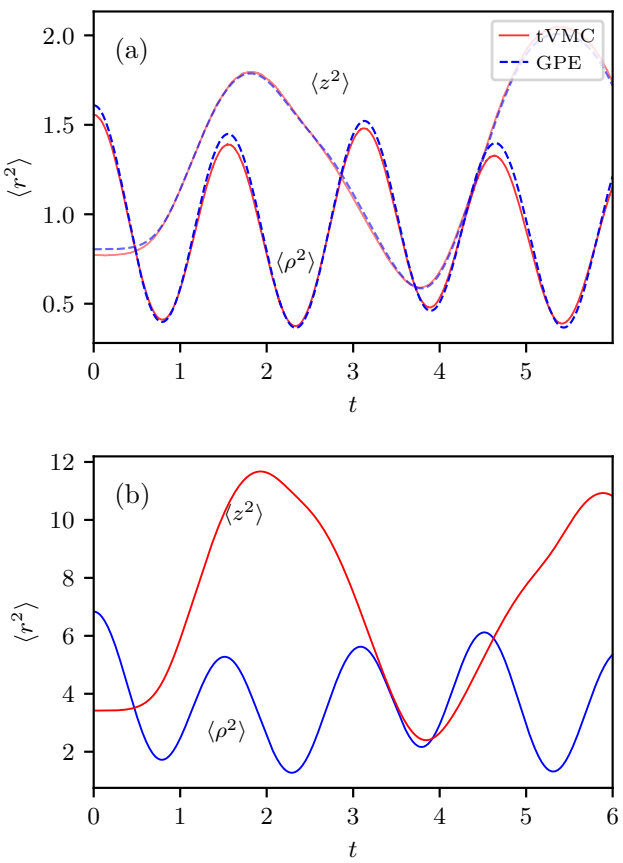

FIG. 4. GPE dynamics from the VMC-derived EOS. (a) N=128 validation: $\langle \rho^2 \rangle(t)$ and $\langle z^2 \rangle(t)$ from frozen-u_2 tVMC (red) and GPE (blue), showing identical breathing frequencies within numerical resolution. (b) N=8192 prediction with the same interaction (C_3=10^{-3}), extrapolating to experimentally relevant system sizes inaccessible to direct tVMC.

treatment, because the bare dipolar integral diverges; the non-trivial correlation hole produced by the shielding repulsion is essential to regularize this integral and fix the coupling. The VMC-derived EOS therefore provides a first-principles route to the effective interaction strength that phenomenological GP theories must take as an external input.

The main quantitative limitation visible here is the few-percent offset in ground-state moments at $N = 128$, which points to gradient corrections beyond the LDA [30]; such terms, of the form $\propto (\nabla\rho)^2/\rho$, could be systematically extracted from VMC at spatially modulated densities and incorporated as a beyond-LDA energy functional. The reduction is expected to break down for quenches faster than the pair-equilibration time ($\tau_{\rm quench} \lesssim \tau_{\rm pair}$), for protocols that nucleate vortices, or near phase transitions (e.g., to a quantum droplet or supersolid) where the pair correlations acquire strong density dependence. Within its domain of validity, the framework extends naturally to other collective modes (scissors, quadrupole), time-dependent trap geometries, and two-component molecular mixtures relevant to current experiments [1, 6, 7].

Conclusion.—We have demonstrated that correlated many-body dynamics from tVMC maps exactly onto superfluid hydrodynamics within the frozen-u_2 manifold, with the required equation of state obtained from a single uniform-system VMC calculation. The resulting generalized GPE reproduces the breathing dynamics of microwave-shielded dipolar bosons from $N = 8$ to $N = 128$ with below-1% frequency error and $\sim$5% amplitude error, and predicts a robust breathing frequency $\omega_{\rm br} \approx 4.19$ that persists from $N = 8$ to $N = 8192$ as the interaction fraction grows from 3.5% to 38%—a prediction directly testable in current molecular BEC experiments. More broadly, the reduction establishes a general strategy for linking correlated variational wave functions to predictive macroscopic dynamics: extract an equation of state from VMC on the uniform system, then evolve via hydrodynamics. This route should be applicable to other quantum fluids with strong short-range correlations, including quantum droplets [10, 16] and dipolar supersolids [8], wherever the pair-correlation sector equilibrates faster than the macroscopic dynamics of interest.

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[1] N. Bigagli, W. Yuan, S. Zhang, B. Bulatovic, T. Karman, I. Stevenson, and S. Will, Observation of Bose–Einstein condensation of dipolar molecules, Nature **631**, 289 (2024).
[2] Z. Shi, Z. Huang, F. Deng, W.-J. Jin, S. Yi, T. Shi, and D. Wang, Bose-Einstein condensate of ultracold sodium-rubidium molecules with tunable dipolar interactions (2025), arXiv:2508.20518 [cond-mat.quant-gas].
[3] L. Anderegg, S. Burchesky, Y. Bao, S. S. Yu, T. Karman, E. Chae, K.-K. Ni, W. Ketterle, and J. M. Doyle, Observation of microwave shielding of ultracold molecules, Science **373**, 779 (2021).
[4] A. Schindewolf, R. Bause, X.-Y. Chen, M. Duda, T. Karman, I. Bloch, and X.-Y. Luo, Evaporation of microwave-shielded polar molecules to quantum degeneracy, Nature **607**, 677 (2022).

[5] X.-Y. Chen, S. Biswas, S. Eppelt, A. Schindewolf, F. Deng, T. Shi, S. Yi, T. A. Hilker, I. Bloch, and X.-Y. Luo, Ultracold field-linked tetratomic molecules, Nature **626**, 283 (2024).

[6] J. Lin, G. Chen, M. Jin, Z. Shi, F. Deng, W. Zhang, G. Quéméner, T. Shi, S. Yi, and D. Wang, Microwave shielding of bosonic NaRb molecules, Phys. Rev. X **13**, 031032 (2023).

[7] M. Duda, X.-Y. Chen, A. Schindewolf, R. Bause, J. von Milczewski, R. Schmidt, I. Bloch, and X.-Y. Luo, Transition from a polaronic condensate to a degenerate Fermi gas of heteronuclear molecules, Nat. Phys. **19**, 720 (2023).

[8] L. Chomaz, I. Ferrier-Barbut, F. Ferlaino, B. Laburthe-Tolra, B. L. Lev, and T. Pfau, Dipolar physics: a review of experiments with magnetic quantum gases, Rep. Prog. Phys. **86**, 026401 (2023).

[9] T. Lahaye, C. Menotti, L. Santos, M. Lewenstein, and T. Pfau, The physics of dipolar bosonic quantum gases, Rep. Prog. Phys. **72**, 126401 (2009).

[10] D. S. Petrov, Quantum mechanical stabilization of a collapsing Bose-Bose mixture, Phys. Rev. Lett. **115**, 155302 (2015).

[11] F. Wächtler and L. Santos, Quantum filaments in dipolar Bose-Einstein condensates, Phys. Rev. A **93**, 061603 (2016).

[12] F. Dalfovo, S. Giorgini, L. P. Pitaevskii, and S. Stringari, Theory of Bose-Einstein condensation in trapped gases, Rev. Mod. Phys. **71**, 463 (1999).

[13] D. S. Jin, J. R. Ensher, M. R. Matthews, C. E. Wieman, and E. A. Cornell, Collective excitations of a Bose-Einstein condensate in a dilute gas, Phys. Rev. Lett. **77**, 420 (1996).

[14] M.-O. Mewes, M. R. Andrews, N. J. van Druten, D. M. Kurn, D. S. Durfee, C. G. Townsend, and W. Ketterle, Collective excitations of a Bose-Einstein condensate in a magnetic trap, Phys. Rev. Lett. **77**, 988 (1996).

[15] A. R. P. Lima and A. Pelster, Quantum fluctuations in dipolar Bose gases, Phys. Rev. A **84**, 041604 (2011).

[16] F. Böttcher, M. Wenzel, J.-N. Schmidt, M. Guo, T. Langen, I. Ferrier-Barbut, T. Pfau, R. Bombín, J. Sánchez-Baena, J. Boronat, and F. Mazzanti, Dilute dipolar quantum droplets beyond the extended Gross-Pitaevskii equation, Phys. Rev. Research **1**, 033088 (2019).

[17] G. Carleo, F. Becca, M. Schiró, and M. Fabrizio, Localization and glassy dynamics of many-body quantum systems, Sci. Rep. **2**, 243 (2012).

[18] G. Carleo, F. Becca, L. Sanchez-Palencia, S. Sorella, and M. Fabrizio, Light-cone effect and supersonic correlations in one- and two-dimensional bosonic superfluids, Phys. Rev. A **89**, 031602 (2014).

[19] G. Carleo, L. Cevolani, L. Sanchez-Palencia, and M. Holzmann, Unitary dynamics of strongly interacting Bose gases with the time-dependent variational Monte Carlo method in continuous space, Phys. Rev. X **7**, 031026 (2017).

[20] F. Deng, X. Hu, W.-J. Jin, S. Yi, and T. Shi, Two- and many-body physics of ultracold molecules dressed by dual microwave fields, Nat. Commun. **16**, 11219 (2025).

[21] R. Jastrow, Many-body problem with strong forces, Phys. Rev. **98**, 1479 (1955).

[22] E. Feenberg, *Theory of Quantum Fluids* (Academic Press, New York, 1969).

[23] See Supplemental Material for details on the basis-function expansion, cusp treatment, TDVP derivation of the hydrodynamic equations, EOS fitting procedure, and numerical convergence tests.

[24] J.-P. Hansen and I. R. McDonald, *Theory of Simple Liquids*, 3rd ed. (Academic Press, 2006).

[25] P. A. M. Dirac, Note on exchange phenomena in the thomas atom, Math. Proc. Cambridge Philos. Soc. **26**, 376 (1930).

[26] A. D. McLachlan, A variational solution of the time-dependent Schrödinger equation, Mol. Phys. **8**, 39 (1964).

[27] L. P. Pitaevskii and S. Stringari, *Bose-Einstein Condensation and Superfluidity*, 2nd ed. (Oxford University Press, 2016).

[28] Y. Castin and R. Dum, Bose-Einstein condensates in time dependent traps, Phys. Rev. Lett. **77**, 5315 (1996).

[29] S. Stringari, Collective excitations of a trapped Bose-condensed gas, Phys. Rev. Lett. **77**, 2360 (1996).

[30] F. Ancilotto, M. Barranco, F. Coppens, J. Eloranta, N. Halberstadt, A. Hernando, D. Mateo, and M. Pi, Density functional theory of doped superfluid liquid helium and nanodroplets, Int. Rev. Phys. Chem. **36**, 621 (2017).

[31] J. Bradbury, R. Frostig, P. Hawkins, M. J. Johnson, C. Leary, D. Maclaurin, G. Necula, A. Paszke, J. VanderPlas, S. Wanderman-Milne, and Q. Zhang, JAX: composable transformations of Python+NumPy programs (2018).